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MA1220 - Calculus II Examination

Date: October 11, 2025 Maximum Marks 47

Time: 120 minutes (10:00 a.m.–12:00 noon)

Q.0. Make an index on page 1. For making an index, you get **(+2 marks)**; for not making an index, you get **(–2 marks)**. (2 marks)

Q.1 Mark each statement True or False (**no justification required**) $(6 \times 1 = 6$ marks)

(a) Suppose $f(x, y)$ is defined on a disk D that contains the point (a, b). If the functions f_{xy} and f_{yx} are both continuous on D, then

$$f_{xy}(a, b) = f_{yx}(a, b).$$

(b) If $f(x, y)$ is continuous and f_x and f_y exist, then for any unit vector $\mathbf{u} = \langle a, b \rangle$,

$$D_{\mathbf{u}} f(x, y) = f_x(x, y)\, a + f_y(x, y)\, b.$$

(c) If $f(x) \geq g(x)$ on the interval $[a, \infty)$ and $\int_a^\infty f(x)\, dx$ converges, then $\int_a^\infty g(x)\, dx$ also converges.

(d) If $f_x(x, y)$ and $f_y(x, y)$ exist and are continuous on an open disk $\mathcal{D}$, then $f(x, y)$ is differentiable on $\mathcal{D}$.

(e) If f is differentiable on $[a, b]$, then the arc length of the curve $y = f(x)$ for $a \leq x \leq b$ is

$$L = \int_a^b \sqrt{1 + \left[f'(x)\right]^2}\, dx.$$

(f) If $f(x, y)$ has a local maximum or minimum at (a, b), then $f_x(a, b) = 0$ and $f_y(a, b) = 0$.

Q.2 Short Answer (Justification required): Answer the following: $(6 \times 1.5 = 9$ marks)

(a) Let $f(x) = 3x^2 - 12x + 12$. Find all $m > 1$ such that the average value of f on $[1, m]$ equals 3.

(b) If $g(2023) = \pi$, assume that g' is continuous, and $\displaystyle\int_{2023}^{2025} g'(x)\, dx = -\frac{\pi}{2}$, find $g(2025)$.

(c) If $f(x, y, z) = \dfrac{xy + yz^2 + xz^2}{x^2 + y^2 + z^4}$ for $(x, y, z) \neq (0, 0, 0)$, does $\displaystyle\lim_{(x,y,z)\to(0,0,0)} f(x, y, z)$ exist?

(d) Determine whether the integral $\displaystyle\int_0^\pi \frac{\sin^2 x}{\sqrt{x}}\, dx$ converges or diverges.

(e) Find an equation of the tangent plane to the surface $z = e^{x-y}$ at the point $(2, 2, 1)$.

(f) Find the area enclosed by the curves $x = \sin y$ and $x = 1 - \cos y$ for $0 \leq y \leq \frac{\pi}{2}$.

Q.3 Answer the following. (14 marks)

(a) Find the area of the region bounded by the graphs of $y = \sin x$ and $y = x\sin(x^2)$ for $0 \le x \le 1$.

(b) Find the volume V obtained by rotating the region enclosed by the curves $y = 2\sqrt{x}$ and $y = x$ about the line $x = -2$. (*Note: Do not evaluate the integral; writing the correct integral expression for V is sufficient.*)

(c) Find the area of the surface obtained by rotating the curve $x = \sqrt{a^2 - y^2}$ for $0 \le y \le \frac{a}{2}$ about the x-axis.

(d) Let $f : [a, b] \to \mathbb{R}$ be continuous and define

$$g(x) = \int_a^x f(t)\,dt \quad (a \le x \le b).$$

Prove that g is differentiable on (a, b).

Q.4 Answer the following. (16 marks)

(a) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(0,0) = 0$ and

$$f(x,y) = \begin{cases} x\sin\dfrac{1}{x} + y\sin\dfrac{1}{y}, & \text{if } x \neq 0 \text{ and } y \neq 0, \\ x\sin\dfrac{1}{x}, & \text{if } x \neq 0 \text{ and } y = 0, \\ y\sin\dfrac{1}{y}, & \text{if } y \neq 0 \text{ and } x = 0. \end{cases}$$

Show that f is continuous at $(0,0)$ (give an ε-δ proof). Show that the partial derivatives of f do not exist at $(0,0)$.

(b) Let $f(x,y) = x^3 + xy^3$. Find all unit directions at the point $(1,1)$ for which the directional derivative is 0.

(c) Let $f : \mathbb{R}^2 \to \mathbb{R}$ be defined by $f(x,y) = x^3$. Find the local maximum and minimum values and the saddle point(s) of f (if they exist).

(d) Find the extreme values of $f(x,y) = 2x + 4y$ subject to the constraint $g(x,y) = x^2 + y^2 - 5 = 0$.